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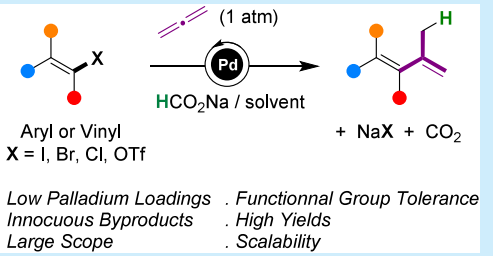
Exploring Pd-Catalyzed Hydrocarbation of Allenes: Focusing on Propa-1,2-diene for the Isopropenylation of C(sp^2) Halides

Louis Perrault, Romain Magny, Lucas Pagès, Rayane Mekerri, Clément Burgevin, Damien Cordeau, and Emmanuel Roulland*

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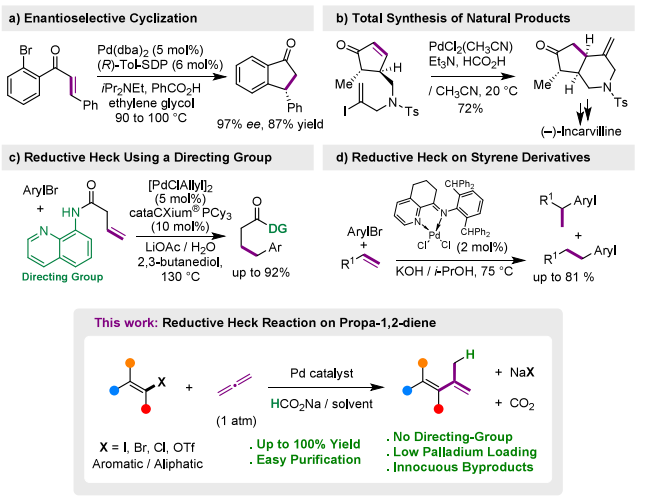
ABSTRACT: The operatively simple Pd-catalyzed allene hydrocarbation described here uses HCO_2Na and produces innocuous CO_2 and sodium halide, and pure coupling products are isolated via water/ether extraction in most cases. This new transformation displays a broad C(sp^2) halide scope, functional group tolerance, and good to excellent yields and allows inexpensive selective deuteration. Pd loadings of 0.10 mol % remain effective, and gram-scale reactions were conducted. The protocol extends to O-, N-, Si-, and C-substituted allenes, demonstrating its versatility.



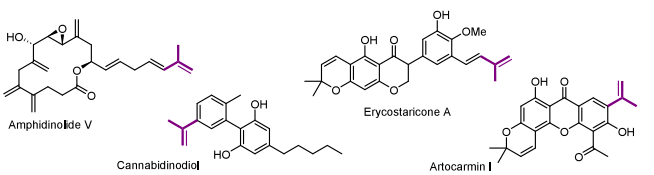
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The Heck coupling is a widely used method enabling C(sp^2) halides to react with diverse alkenes under Pd catalysis mostly.[1] The catalytic cycle involves a β-hydride elimination, restoring an unsaturation, but if a hydride donor is added, concurrent hydride transfer occurs, leading to saturated products. This hydrocarbation or reductive Heck allows enantioselective cyclization (Scheme 1a);[2] in total syntheses,[3] it permitted ring closures of (−)-incarvilline (Scheme 1b),[4] sesterstatins,[5] menogaril, and nogalamycin.[6] Sometimes, competing dehalogenation prevails over carbopalladation as in the case of the synthesis of (±)-aspidophylline A.[7] However, competing β-hydride elimination, dehalogenation, and regioselectivity issues make the intermolecular reductive Heck approach challenging. Efficient and regioselective transformations were achieved however with alkenes bearing Pd-chelating groups (Scheme 1c),[8] styrenes[9] (Scheme 1d), norbornenes,[10] allyl alcohols,[11] and alkynes,[12] but unbiased alkenes generally exhibit incomplete regioselectivity.[9,11] Our previous study on allene−alkyne cross-coupling[13] led us to appreciate the rich reactivity of allenes and their utility in total synthesis.[14,15] Allenes are competent in Heck reactions;[16−18] however, reductive Heck of allenes is notably absent from the literature. Here, we explore this uncharted territory and show that prop-1,2-diene enables a practical method of isopropenylation of C(sp^2) halides. Since more than 16 000 prop-1-en-2-yl-containing molecules are described, notably natural products[19−22] (Scheme 2) and pharmaceutical drugs, new methods for introducing the prop-1-en-2-yl group from C(sp^2) halides are highly desirable, especially if they are more atom-economical or environmentally friendly than traditional cross-couplings (Suzuki−Miyaura, Stille, or Negishi). First,

Scheme 1. Reductive Heck, Applications, and Our Work



Scheme 2. Some Prop-1-en-2-yl-Bearing Natural Products



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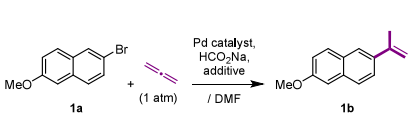


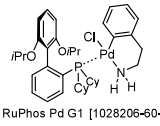
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we imagined that Heck–Jeffery-like conditions could be implemented[23] by simply adding HCO_2Na as a hydride donor. The reaction was carried out at 60 °C under 1.0 bar of propa-1,2-diene, with HCO_2Na (1.2 equiv), $Pd_2(dba)_3$ (3 mol %), and Bu_4NBr (1.0 equiv), in degassed dry DMF, and 2-bromo-6-methoxynaphthalene **1a** was used as the model substrate (Table 1, entry 1). After 18 h, the ^1H NMR

Table 1. First Trials of Reductive Heck on Propa-1,2-diene[a]





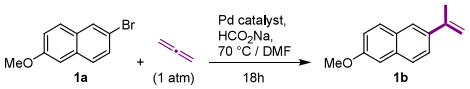
RuPhos Pd G1 [1028206-60-1]

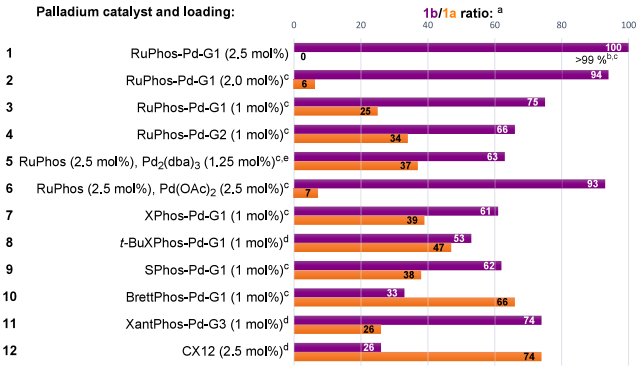
	catalyst	HCO_2Na (equiv)	additive	conditions: time, temperature, and **1b**/**1a** ratio[b]
1	$Pd_2(dba)_3$ 3 mol %	1.2	Bu_4NBr, 1.0 equiv	18 h, 60 °C 20:80[d]
2	$Pd_2(dba)_3$ 3 mol %	3.0	Bu_4NBr, 3.0 equiv	18 h, 60 °C 43:57[d]
3	$PdCl_2(dppf)$ 5 mol %	3.0	no	18 h, 70 °C 74:26[d]
4	RuPhos-Pd-G1 2.5 mol %	3.0	no	18 h, 70 °C 100:0, >99%[c,d]

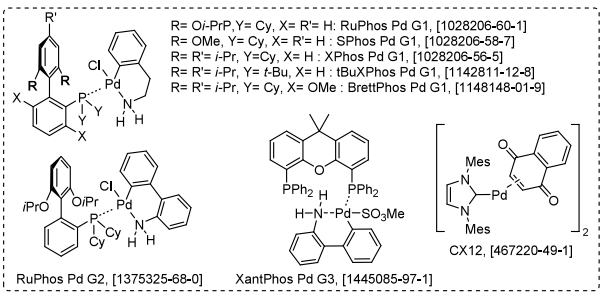
[a]With 0.5 mmol of **1a** in degassed DMF. [b]From the ^1H NMR experiment on the crude reaction mixture. [c]Isolated yield. [d]No byproduct detected.

spectrum shown a clean crude reaction mixture made of **1a** and expected 2-methoxy-6-(prop-1-en-2-yl)-naphthalene **1b** in an unfavorable 80:20 ratio. This initial encouraging result demonstrated the feasibility of reductive Heck on an allene, moreover, in an intermolecular fashion. The absence of detectable byproducts hinted at the possibility of effective improvements. Increasing the amounts of HCO_2Na and Bu_4NBr resulted in a 43:57 **1b**/**1a** ratio (entry 2). However, catalyst degradation into Pd black limited conversion; therefore, stabilizing phosphine ligands were used. $PdCl_2(dppf)$ (entry 3) gave a favorable 74:26 **1b**/**1a** ratio, even if Pd black formation hindered reaching completion. Highly active RuPhos-Pd-G1[24] was then employed (entry 4), resulting in a spectacular improvement; only **1b** was detected in a very clean crude reaction mixture, achieving a quantitative yield. Lower RuPhos-Pd-G1 catalyst loadings were tested (Table 2, entries 2 and 3), with 2 mol % completion being no longer reached over 18 h at 70 °C, and 1 mol % gave a 75:25 **1b**/**1a** ratio. With incomplete conversion of **1a** being shown, these "at the limit conditions" were harnessed to detect an eventual more efficient catalyst. Other RuPhos-based systems, such as RuPhos-Pd-G2 or RuPhos/$Pd_2(dba)_3$ (Table 2, entries 4 and 5), gave similar ≈65:35 **1b**/**1a** ratios. The $Pd(OAc)_2$/RuPhos combination gave results similar to those of reference conditions, featuring a cheaper alternative (entry 6). XPhos-Pd-G1, *t*-BuXPhos-Pd-G1, and SPhos-Pd-G1 (entries 7–9) yielded similar **1b**/**1a** ratios from 60:40 to 50:50. With BrettPhos-Pd-G1, the **1b**/**1a** ratio dropped to 33:66 (entry 10), while XantPhos-Pd-G3 gave results similar to reference conditions (entry 11). Carbene-liganded Pd catalyst CX12 allowed only partial conversion with a 26:74 **1b**/**1a** ratio, even at 5 mol % (entry 12), with unidentified byproducts also forming. This series of experiments led us to retain RuPhos-Pd-G1 to further explore this transformation.

Table 2. Palladium Ligand Optimization



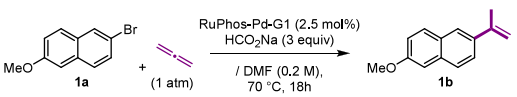


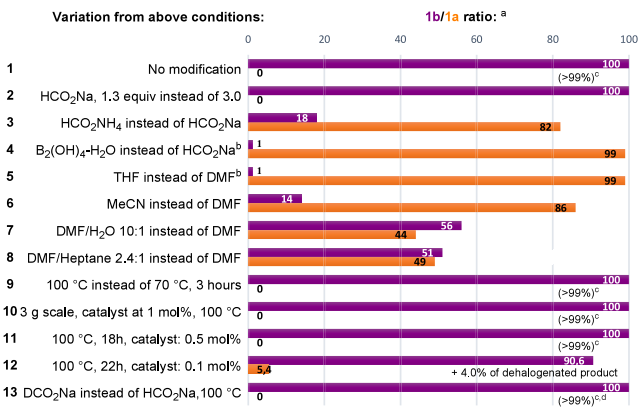


[a]Determined by ^1H NMR on the crude reaction mixture. [b]Isolated yield. [c]No byproduct detected. [d]Non-defined byproducts formed. [e]Pd black formation.

Variation around the hydride source was explored: Using 1.3 equiv of HCO_2Na instead of 3.0 still allowed completion to be reached (Table 3, entry 2). Replacing HCO_2Na by HCO_2NH_4 or the water–$B_2(OH)_4$ couple[25] proved unsatisfactory (Table 3, entries 3 and 4). Replacing DMF by THF or MeCN or adding heptane or water to DMF to tune propa-1,2-diene

Table 3. Variation around Established Efficient Conditions





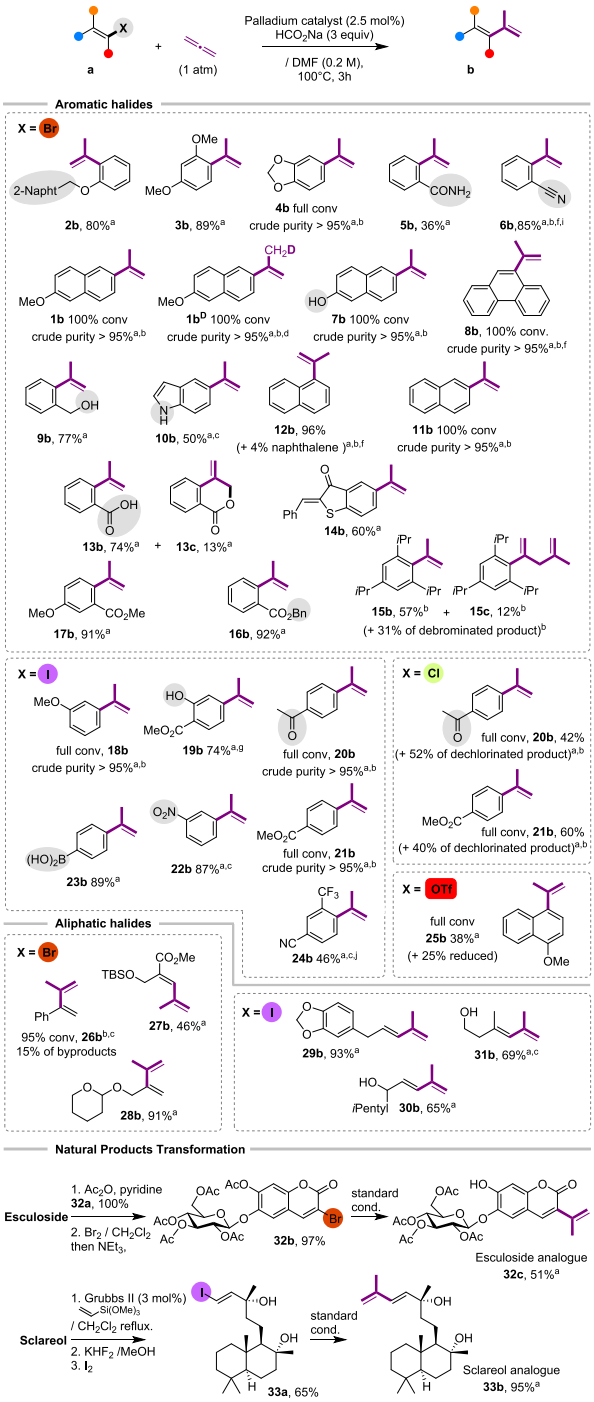
[a]Determined by ^1H NMR on the crude reaction mixture. [b]Carried out on 3,4-methylidenedioxy-bromobenzene. [c]Isolated yield. [d]A >99% D incorporation at the allylic position of **1b**^{D}.

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solubility (entries 5−8) gave lower conversions. At 100 °C, the reaction reaches completion in less than 3 h (entry 9) even on a 2.00 g scale at 1.0 mol % RuPhos-Pd-G1 (entry 10). Full conversion was maintained at 0.5 mol % (240 mg scale, 18 h, entry 11), and 95% conversion was reached at 0.1 mol % on a 473 mg scale over 22 h (entry 12), underscoring the efficiency of the reaction. With DCO_2Na (entry 13), regioselective deuteration at the allylic methyl group confirmed that the formate anion is indeed the source of hydride and that no hydrogen atom scrambling occurred. This provides an easy and cost-effective approach to accessing deuterated molecules, which hold significant practical value, for instance, in medicinal applications.[26] Note that compound **1b** is a potential synthetic intermediate toward naproxene.[27]

We explored variations around the nature of aryl halide and extended the study to vinyl halides using the following conditions: propa-1,2-diene (1 bar), RuPhos-Pd-G1 (2.5 mol %), and HCO_2Na (3.0 equiv) in DMF at 100 °C. Electron-rich methoxy-substituent aryl bromides gave desired products **1b**, **2b**, **3b**, and **4b** (Scheme 3) efficiently. Non-protected phenol **7a** gave **7b** quantitatively. While 2-bromonaphthalene **11a** gave **11b** quantitatively, 1-bromonaphthalene **12a** led to a 76:24 **12b**/naphthalene mixture. However, replacing RuPhos by XantPhos decreased dehalogenation to 5%, but 1-allylnaphthalene formed (6%), issuing from an alternative carbopalladation regioselectivity. With 2-bromobenzonitrile **6a**, expected **6b** was obtained along with 20% of benzonitrile, XantPhos suppressed dehalogenation, but 2-allylbenzonitrile formed (15%). Hindered **15a** achieved partial conversion into **15b** accompanied with the corresponding dehalogenated product, and **15c** featuring an unexpected double propenylation. Non-protected alcohol has no impact as **9b** formed in a good yield without the corresponding cyclization product, while competing cyclization took place with **13a**, featuring a carboxylic acid adjacent to the leaving halogen, proving the existence of a π-allyl palladium intermediate. Free amide in 2-bromobenzamide **5a** led to no lactam, and only amide **5b** was isolated. These reductive conditions are mild and selective enough to leave benzyl or 2-naphthyl protecting groups or even nitro groups intact, with **16b**, **2b**, and **22b** forming in good yields. With 5-bromoindol **10a**, the reaction gave mediocre yields of **10b**. Aurone **14b** could be obtained from **14a**[28] in 60% yield, proving that the sulfur atom is tolerated. Aryl iodides proved competent as well. Electron-rich **18b** was isolated in a high yield, and phenol **19b** was obtained in a yield of 74% but remained contaminated by 5% of the dehalogenation byproduct. With boronic acids remaining inert toward cross-coupling under pH-neutral conditions, (4-iodophenyl)boronic acid **23a** could be selectively transformed into **23b**. Ketones proved compatible, with **20b** forming quantitatively. With iodinated substrates, Pd black formation was never observed, with iodide anions apparently acting as catalyst protection. We exploited this protective effect with bromophenanthrene **8a**, which without NaI addition, use to give a 64:36 mixture of **8b** and phenanthrene **8c** with rapid Pd black formation. Adding 1 equiv of NaI delayed catalyst degradation but only slightly improved the **8b**/**8c** ratio to 76:24. Replacing RuPhos with XantPhos completely suppressed dehalogenation, yielding **8b** quantitatively. Electro-poor derivative **24a** gave a 66:33 ratio of **24b**/deiodinated **24c**, and XantPhos allowed no improvement; however, using less HCO_2Na (2 equiv) under higher propa-1,2-diene pressure (2−3 bar) allowed the ratio to increase to 80:20. Aryl chlorides reacted quickly but with competing dehalogenation, with ketone **20b** and ester **21b** being contaminated with ≈50% of dehalogenated products. With **Chloro-20a** using RuPhos with NaI, no conversion occurred, while **Iodo-20a** reacted cleanly and quantitatively. **Chloro-20a** proved poorly reactive by using XantPhos (≈1% of **20b** formed). Similar observations were made with **Chloro-21a**.

Scheme 3. Substrate Scope Exploration



[a]Isolated yield. [b]Determined by ^1H NMR. [c]Volatile. [d]DCO_2Na instead of HCO_2Na. [e]A total of 1 equiv of NaI was added. [f]XantPhos-Pd-G3 instead of RuPhos-Pd-G1. [g]Contaminated with 5% methyl salicylate. [h]Contaminated with 5% methyl 3-methoxybenzoate. [i]Contaminated with 15% 2-allylbenzonitrile. [j]HCO_2Na (2 equiv) and propa-1,2-diene (2−3 bar).

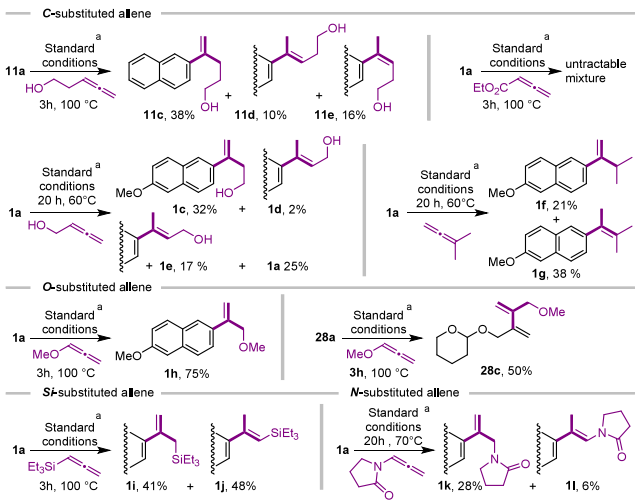
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Aryl triflates proved reactive under our conditions but also with important dehalogenation levels, even with XantPhos and NaI, with **25b** coming along with 45% of the dehalogenated product.

Vinyl halides proved to be competent substrates, giving dienes with configuration retention. Compounds **29a**, **30a**, and **31a** were prepared according to a one-pot three-step procedure developed in the lab.[15] Functionalization of the complex structures of natural products proved to be efficient. Brominated esculoside **32b**, a glycosylated coumarin, was converted into its analogue **32d**, and iodinated sclareol **33a**, a diterpene, was transformed into diene **33b** in 95% yield.

Substituted allenes also proved competent, highlighting the versatility of our method, albeit regioselectivity issues appeared. C-substituted allene hexa-4,5-dien-1-ol (Scheme 4)

Scheme 4. Examples Involving Substituted Allenes

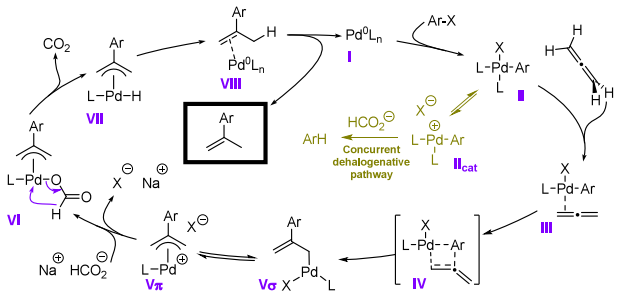


[a]RuPhos-Pd-G1 (2.5 mol %) and HCO_2Na/DMF.

reacted with **11a**, affording a mixture of *exo*-methylene **11c** along with *E* and *Z endo* derivatives **11d** and **11e** (≈50:50, 63%). Bromide **1a** reacted with buta-2,3-dien-1-ol, giving isomers **1c**, **1d**, and **1e**. Congested 3-methylbuta-1,2-diene reacted with **1a**, giving isomers **1f** and **1g** and the dehalogenation product. Methoxyallene, reacted with **1a**, giving *exo*-methylene isomer **1h** selectively (75% yield). With the opposite electronic effect displayed, ethyl buta-2,3-dienoate gave an intractable reaction mixture. Si-substituted allenes proved efficient coupling partners, as with **1a**, triethyl(propa-1,2-dien-1-yl)silane gave inseparable **1i** and **1j** (≈45:55, 89%). N-substituted allenes, such as 1-(propa-1,2-dien-1-yl)-pyrrolidin-2-one,[29] are possible substrates, with **1k** being isolated in 28% yield along with its enamide isomer **1l** (6%) and the dehalogenation product. The regioselectivity observed with the substituted allenes studied here appears to be mainly governed by steric factors, as in the case of products **11c**–**11e**, **1c**–**1e**, **1f**, **1g**, **1i**, and **1j**. In contrast, O- and N-substituted allenes, such as 1-methoxypropa-1,2-diene and 1-(propa-1,2-dien-1-yl)pyrrolidin-2-one, display complete or nearly complete regioselectivity, which can be attributed to electronic effects. These encouraging results open new possibilities for the transformation of substituted allenes. Further investigations will focus on improving regioselectivity and minimizing the competing dehalogenation pathway.

In a plausible mechanism (Scheme 5), the oxidative addition of Pd^0L_n **I** leads to Pd^{II} **II**, and then allene interacts with this *π*-

Scheme 5. Proposed Reaction Mechanism



philic complex, which through **IV** leads to *σ*-allyl−Pd complex $\mathbf{V}_{\sigma}$ (in equilibrium with cationic *π*-allyl−Pd $\mathbf{V}_{\pi}$). The formate anion then bonds Pd^{II} of $\mathbf{V}_{\pi}$, giving **VI**, within which hydride transfer occurs releasing CO_2[30] and hydrido *π*-allyl−Pd **VII**. Final reductive elimination regenerated Pd^0L_n and released the product. Selective deuteriation experiments leading to $\mathbf{1b}^{\mathbf{D}}$ (Scheme 3) and the formation of lactone **13c** from carboxylate **13a** both support this mechanism.

To conclude, we reported the first example of a reductive Heck reaction involving allenes. The efficiency of this transformation stems from the strong affinity of allenes for *π*-philic Pd^{II} complexes, which in conjunction with the Ruphos ligand suppresses the competing dehalogenation pathway in most cases. Moreover, the use of wide-bite-angle bisphosphine XantPhos effectively prevents recalcitrant dehalogenation with certain substrates, providing valuable insight into strategies for suppressing this side reaction. Using propa-1,2-diene, we developed a mild, high-yielding method to install a prop-1-en-2-yl group across 35 examples, including multigram-scale reactions and natural product functionalization. Both aryl and vinyl bromides and iodides are competent, but with aryl triflates and chlorides, dehalogenation could not be suppressed. Deuterated analogues are easily accessible, and C-, O-, N-, and Si-substituted allenes also proved competent coupling partners. By exploring uncharted territory, this study further enriches the synthetic role of allenes, expands the Pd catalysis scope, and provides a new strategy for the construction of complex molecules.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c03623.

Experimental procedures, spectroscopic data, and copies of ^1H and ^{13}C NMR spectra (PDF)

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Author Contributions

All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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